CS 147: Using the Synthesizer

March 9, 2026

1 Overview

In this class we will do some very simple synthesis of your designs. The primary goal of this exercise is to get a sense for the actual hardware your Verilog is creating. Synthesizing your design will allow us to see:

- The actual gates
- Area
- Delay
- The actual longest path in your design :)

There is a lot more to synthesis optimizations than what we will cover in this class. We will be using Synopsys DC Compiler and a 45nm gate library provided by FreePDK. A lot of the details will be abstracted away and you will be using a simple script called synth.pl which will do most of the work for you.

You should connect to the VPN for logging onto the coe-ee machines. Use RDP to connect to the machines. The instructions are in the Git repo at:

```
assignments/tools/synth/sjsu_remote_cad_machine_access.pdf
```

2 First Time Setup

In the Git repo for the course, you will find the following folder: `assignments/tools/synth`. From this folder, copy the setup file to your home directory on one of the `coe-ee` machine. The setup files you need are:

```
prompt> ls -a assignments/tools/synth/.synopsys_dc.setup
prompt> assignments/tools/synth/synth.pl
```

Upload these two files to a Google drive and then copy them over to your home directory on the CAD machines.

Note: There are about 50 or so machines in the CAD lab. Pick a machine that is lightly loaded for your work.

3 Running the Synthesizer

On an on-going basis, whenever you need to run the synthesizer, the first step is to copy over the associated files. For instance, to run the synthesizer on the Verilog files from HW2.2 (i.e., the ALU), do the following:

1. Copy the Verilog files. Again, copy them to Google drive and then download on the CAD machines. Create a folder `code` in your home directory where you will copy the files.

   ```
   prompt> ls -l coe-ee-cad15.sjsuad.sjsu.edu:code
   ```

2. Use the RDP instructions provided above to access the machines. Once you are logged in, run the following commands to get the environment setup:

   ```
   coe-ee-cad15> csh

   coe-ee-cad15>  source /apps/design_environment.csh
   ```

3. Make a list of Verilog files that are part of the design. Create a text file (named `list.txt` in this folder (`code/hw02/hw2_2`)) with one Verilog filename per line. Additionally,

   - No testbenches should be included in this list
   - No _hier files which are wrappers should be in this list
   - clkrst.v should NOT be in the list, since we don't want to synthesize the clock generator

   For example `list.txt` with the following contents.

   ```
   alu.v
   4_1mux.v
   CLA.v
   ```

4. Check the design.

   ```
   coe-ee-cad15>  cd code/hw02/hw2_2
   coe-ee-cad15> perl ~/synth.pl --list=list.txt --type=other --cmd=check --top=alu
   ```

   Look at the output on the screen and `synth.log`. If there are no errors, go to the next step.

5. Run the synthesis.

   ```
   coe-ee-cad15> perl ~/synth.pl --list=list.txt --type=other --cmd=synth --top=alu
   ```

   This step usually takes a few minutes. The output is in `synth/` folder. You will find an area report, cell report, and a timing report.

6. Optimizing your design. This is analogous to using the GCC compiler to optimize code generation.

   ```
   coe-ee-cad15> perl ~/synth.pl --list=list.txt --type=other --cmd=synth --top=alu --opt
   ```

4 Interpreting the Output Files

Please see this page about how to interpret the outputs in the files generated by the DC compiler.

5 Acknowledgements

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